

NOAH S. PETTINATO

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Open to Relocation

EDUCATION

OREGON STATE UNIVERSITY, Department of Mathematics, Corvallis, OR

June 2025

B.S. in Applied and Computational Mathematics; Minor in Computer Science

Relevant Coursework: *Advanced Calculus I & II, Linear Algebra I & II, Probability, Numerical Analysis, Numerical Partial Differential Equations, Applied Ordinary Differential Equations, Mathematical Modeling, Matrix Algebra, Data Structures, Algorithms, Web Development, Discrete Mathematics*

TECHNOLOGY

Programming and Scripting Languages: Python, MATLAB, SQL, Java, JavaScript, HTML, R, C#

Data and Statistical Tools: NumPy, Pandas, Tableau, Power BI, SSRS, Microsoft Excel (formulas, macros), Crystal Reports

Development and Collaboration Tools: VS Code, GitHub, Git, LaTeX, Salesforce, Playwright

Technical Skills: ETL pipelines, workflow automation, Power Automate, web scraping, AWS, data visualization, dashboard creation, AI training and research, dataset curation, model tuning, runtime benchmarking, solution validation

PROFESSIONAL EXPERIENCE

Axos Bank, Jr. Data Analyst, San Diego, CA

Jan 2025 – Present

- Develop and optimize SQL-driven reporting and business intelligence solutions in SSRS, Tableau, and Excel, partnering with stakeholders to support commercial lending operations.
- Design automated reporting and monitoring workflows using Power Automate, SQL Server, and Python, improving operational efficiency and access to critical business information.
- Built a natural-language data discovery platform integrating Collibra metadata and relationship mapping to accelerate data sourcing and report development across enterprise systems.

MERCOR, Mathematics Expert, (Remote Part-Time, Contract)

Sept 2025 – Present

- AI training and research: Curate and author advanced undergraduate and master's-level mathematics problems, evaluate and refine AI-generated solutions for rigor and clarity, and provide subject matter expertise to strengthen datasets powering next-generation large language models.

WAGNER ENGINEERING, Data Operations Intern, (Remote Part-Time)

Oct 2025 – Jan 2026

- Automated extraction, cleaning, and structuring of Texas SmartBuy procurement data with Python, Playwright, and pandas. Developed cloud ETL pipelines integrating web scraping and data transformation for workflow automation.

OREGON STATE UNIVERSITY, Undergraduate Research Assistant, Mathematics Dept., Corvallis, OR

April – July 2025

- Developed and implemented nonlinear PDE solvers in MATLAB using Newton methods with analytic Jacobians, adaptive damping, and finite difference schemes for steady-state and time-dependent problems with flux-based Robin boundary conditions.
- Modeled radiative heat transfer and evaluated solver performance through convergence analysis and data mining of simulation outputs, documenting algorithms, assumptions, and validation results.

PETTINATO & ASSOCIATES, Data and Research Intern (Seasonal), Sacramento, CA

June 2019 – July 2025

- Extracted, transformed, and validated claim data using SQL Server, relational queries, and Crystal Reports, supporting reporting, workflow efficiency, and asset recovery operations.

CLASS PROJECTS

NUMERICAL SOLVER FOR HEAT EQUATION, Numerical PDEs

Spring 2025

Developed a MATLAB solver for transient heat conduction using finite difference methods, validating accuracy through test-case comparisons and thermal diffusion analysis.

ALGORITHMIC SOLVER SUITE, Analysis of Algorithms

Winter 2024

Built Python solutions using recursive, greedy, dynamic programming, and machine learning techniques, evaluating performance through complexity analysis and runtime benchmarking.

INDEPENDENT PROJECTS

AAPL STOCK FORECASTING WITH ARIMA

Spring 2025

Forecasted AAPL stock prices using Python and ARIMA on 7 years of historical financial market data, tuning models via AIC/BIC and residual diagnostics. Compared forecasts to actual prices and analyzed key statistics (mean, median, mode) for accuracy. Built interactive dashboards in Tableau, Power BI, and Excel with automated macros and preprocessing.

MATE INTERNATIONAL ROV COMPETITION

2015 – 2016

Led development of an underwater ROV that placed 2nd internationally among 30+ teams, integrating real-time sensor-actuator feedback and custom control systems while directing testing and performance optimization.